

### Packet Capture Tools

**Wireshark** – Network packet analyzer used to capture and inspect network traffic in real time or from saved capture files. Commonly used for network troubleshooting, traffic analysis, and security investigations.

- **Promiscuous Mode** – Network interface mode that allows a device to capture all packets it can see on the network segment, not just packets addressed to its own MAC address.

**TCPdump** – Command-line packet capture tool used to capture and analyze network traffic directly from a terminal. Often used on Linux/Unix systems and servers where a GUI is not available.

- **Command-line (CLI) based tool**
- **tcpdump -i <interface>** – Specifies the network interface to capture traffic from (example: eth0, wlan0).

### Log Analysis Tools / Correlation

**Security Information and Event Management (SIEM)** – a centralized platform that collects, aggregates, correlates, and analyzes log data from multiple sources (servers, endpoints, network devices, applications) to detect threats, support incident response, and meet compliance requirements.

- **Aggregation** – Process of collecting and normalizing log data from multiple sources into a single system for analysis.
- **Correlation** – Process of linking related events across different data sources to identify patterns, anomalies, or potential security incidents.
- **Alerting based on triggers** – Automated notifications generated when predefined rules, thresholds, or conditions are met (e.g., multiple failed logins, malware detection).
- **Visibility (Dashboard)** – Centralized visual view of security events, alerts, trends, and system health to improve situational awareness.
- **Compliance** – Ability to support regulatory and organizational requirements (e.g., log retention, auditing, reporting) such as PCI DSS, HIPAA, or GDPR.
- **Data Retention** – Configuration that defines how long logs and event data are stored to support investigations, forensics, and compliance.

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**Security Orchestration, Automation, and Response (SOAR)** – a platform that automates and coordinates security workflows, integrates multiple tools, and streamlines incident response using predefined actions and playbooks.

- **Automate Routine Tasks** – Automatically performs repetitive actions such as ticket creation, IP blocking, user isolation, or malware containment.
- **Playbook** – Predefined, step-by-step workflow that guides or automates responses to specific incident types.
- **Notification** – Automated alerts or messages sent to analysts or systems when actions are taken or incidents occur.

## Endpoint Security

**Endpoint Detection and Response (EDR)** – an endpoint security solution that continuously monitors endpoint activity to detect, investigate, and respond to threats in real time.

- **Detects and Analyzes Malicious Activity** – Monitors endpoint behavior to identify malware, exploits, lateral movement, and suspicious processes.
- **Improves Incident Response** – Provides visibility, investigation tools, and response actions to reduce detection and containment time.
- **Proactive Prevention** – Uses behavioral analysis and threat intelligence to stop attacks before full compromise occurs.
- **Risk Assessment** – Evaluates endpoint activity to identify vulnerabilities, exposure, and the impact of threats.
- **Incident Investigation** – Enables analysts to trace attack timelines, root cause, and attacker behavior on endpoints.
- **URL Filter** – Controls or blocks access to malicious or unauthorized websites based on categories or threat intelligence.
- **Honeypot** – A decoy system or resource designed to attract attackers and collect intelligence on attack techniques.

## DNS & IP Tools

**Domain Name Service (DNS) & Internet Protocol (IP)** – Core networking technologies where DNS resolves domain names to IP addresses, and IP provides logical addressing and routing to deliver packets across networks.

- **WHOIS** – Query protocol and database used to retrieve registration details for domains or IP addresses (registrar, ownership, contact info, registration dates).
- **AbuseIPDB** – Threat intelligence database that tracks and reports malicious IP addresses based on community and provider submissions.
- **Domain Dossier (CentralOps)** – Investigation tool that aggregates WHOIS, DNS records, network information, and domain relationships for threat analysis. <https://centralops.net/co/domaindossier>

### File Analysis Tools

**Strings** – A file analysis utility that extracts human-readable text strings from binary files to help identify embedded commands, URLs, file paths, error messages, or malware indicators.

- **Common flags:**
  - **-a** – Scan the entire file (not just initialized data sections)
  - **-n** – Set minimum string length to display (default is usually 4)
  - **-t x | d | o** – Display offsets in hexadecimal, decimal, or octal
  - **-e l | b** – Specify character encoding (little-endian or big-endian Unicode)

**VirusTotal** – Online malware analysis service that scans files, hashes, URLs, IP addresses, and domains using multiple antivirus engines and threat intelligence sources to assess maliciousness.

### Sandbox Tools

**Joe Sandbox** – Cloud-based malware analysis sandbox that executes suspicious files and URLs in an isolated environment to observe behavior, indicators of compromise (IOCs), and threat characteristics.

**Cuckoo Sandbox** – Open-source automated malware analysis sandbox used to detonate files in a controlled environment and collect behavioral data such as network activity, file changes, and registry modifications.

### Common Detection Techniques

**Pattern Recognition** – A detection technique that identifies known malicious behaviors or characteristics by comparing activity against established signatures, baselines, or behavioral patterns.

- **Command and Control (C2) Communications** – Network traffic used by attackers to remotely control compromised systems, issue commands, and exfiltrate data.

**Interpreting Suspicious Commands** – Analysis of executed commands or scripts to identify malicious intent, attacker techniques, or post-exploitation activity.

**File Analysis** – Examination of files to determine maliciousness using static and dynamic techniques such as metadata review, string extraction, and behavioral execution.

- **Hashing** – Process of generating a fixed-length value from a file to uniquely identify it and verify integrity or compare against known malicious hashes.

### Email Analysis

**Header** – Metadata within an email that shows routing, authentication results, and transmission details used to assess legitimacy.

- **Display From** – The sender name shown to the user, which can be spoofed and does not guarantee authenticity.
- **Envelope From** – The actual sender address used during SMTP transmission and mail routing, often checked during authentication.
- **Received From/By** – Header fields that document the path an email took through mail servers, useful for tracing origin.

**Impersonation** – A social engineering technique where an attacker pretends to be a trusted entity to trick recipients into taking harmful actions.

**Embedded Links** – URLs embedded in email content that may redirect users to malicious or credential-harvesting websites.

### DKIM, DMARC, SPF

**Domain Keys Identified Mail (DKIM)** – Email authentication method that uses cryptographic signatures to verify that a message was not altered in transit and was sent by an authorized domain.

- **Digital Signatures** – Cryptographic hashes added to email headers to verify message integrity and sender authenticity.
  - **Public/Private Keys** – Asymmetric key pairs where the private key signs the message and the public key is used to verify the signature.
- Digital Signatures- Please define this, ChatGPT.
  - Public/Private Keys - Please define this, ChatGPT.

**Domain-based Message Authentication, Reporting, and Conformance (DMARC)** – Email authentication policy that builds on SPF and DKIM to define how receivers should handle failed authentication and provide reporting.

- **Policy-driven control** – Rules that specify how email failures are handled.
  - **Quarantine** – Suspicious messages are delivered to spam or quarantine folders.
  - **Drop/Reject** – Messages that fail authentication are rejected outright.
    - **-all** = reject all
  - **Deliver** – Messages are delivered normally despite authentication results.
    - **+all** = accept all
  - **Flag** – Messages are marked or monitored but still delivered.
    - **~all** = flag all

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**Sender Policy Framework (SPF)** – Email authentication protocol that specifies which mail servers are authorized to send email on behalf of a domain.

### User Behavior Analysis

**Abnormal Account Activity** – User behavior analysis indicator involving actions that deviate from normal patterns, potentially signaling compromise.

- Evidence of communications with known bad IPs or domains
- Encrypted traffic between a device with a remote host
- Beacon
- Port associated with C2

**Impossible Travel** – Detection scenario where user logins occur from geographically distant locations within an unrealistically short time frame.

- Time Zone
- Geo Location

### Programming Languages

**JavaScript Object Notation (JSON)** – Lightweight data-interchange format commonly used for structured data exchange between applications.

**Extensible Markup Language (XML)** – Markup language used to store and transport structured data with customizable tags.

**Python** – High-level programming language widely used in cybersecurity for scripting, automation, data analysis, and tool development.

**Shell Script** – Script written for a command-line shell to automate system tasks and administrative operations.

**Regular Expressions** – Pattern-matching syntax used to search, match, and manipulate text strings.